

Digital Design (SET 111)

L05

Logic Gates

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- ☐ The NOT Gate
- ☐ The AND Gate
- ☐ The OR Gate
- ☐ The NAND Gate
- ☐ The NOR Gate
- ☐ The XOR Gate
- ☐ The XNOR Gate



- ☐ **The NOT Gate**
- ☐ **The AND Gate**
- ☐ **The OR Gate**
- ☐ **The NAND Gate**
- ☐ **The NOR Gate**
- ☐ **The XOR Gate**
- ☐ **The XNOR Gate**

The NOT Gate(1/2)

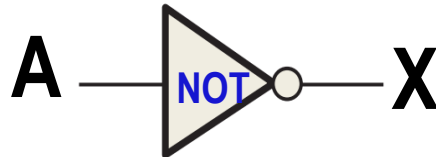


□ Definition

- The NOT Gate performs the operation called **inversion** or **complementation**. $1 \longrightarrow 0$
- The NOT Gate changes **one logic** level to the **opposite** level ,In terms of bits. $0 \longrightarrow 1$

□ Symbol

- **Standard** logic symbols for the inverter (ANSI/IEEE Std. 91-1984/ Std. 91a-1991)



□ NOT Gate Truth Table

- Truth table is a combinations of binary **input** and **output** variables
- # of rows of truth table = 2^n , where **n**=# of input variables
- Ex. **n=1** $\rightarrow$ # of rows = $2^1=2$

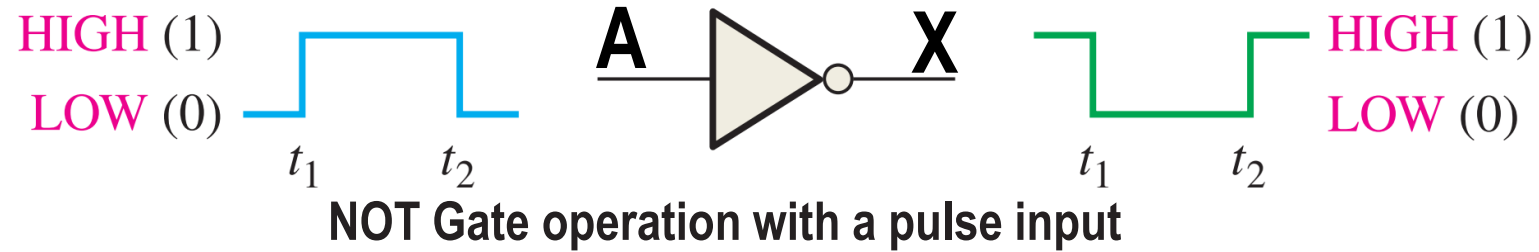
Truth Table

Input	Output
A	X
0	1
1	0

The NOT Gate (2/2)



NOT Gate Operation

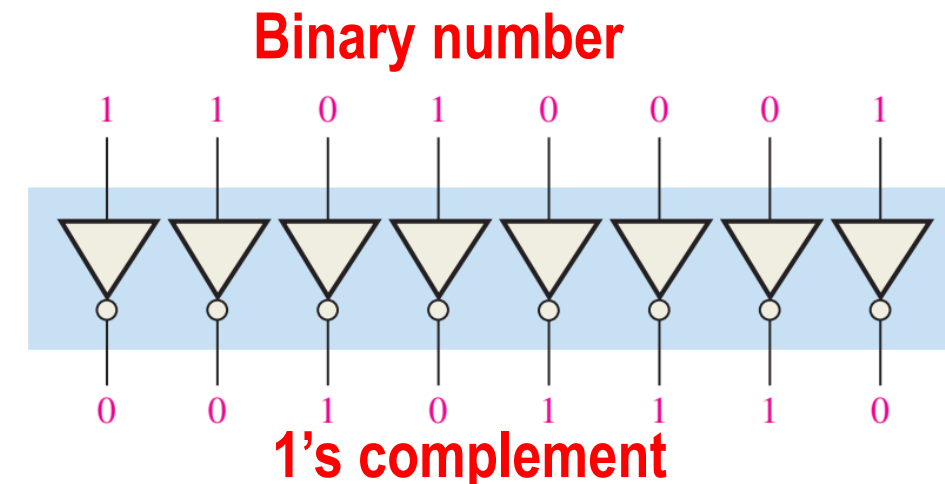


NOT Gate Logic Expression

- The complement of a variable is designated by a **bar** over the letter
- If the input variable is called A and the output variable is called X $\rightarrow X = \bar{A}$

Application

- 1's complement circuit using inverters.





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The AND Gate (1/3)

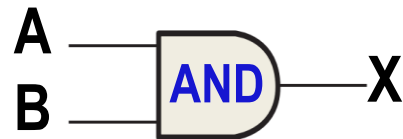


□ Definition

- The **AND** function produces a **HIGH** output only when **all** the inputs are **HIGH**
- Basic **AND** Gate is composed of **two** or more inputs and a single output.

□ Symbol

- Standard logic symbols for the AND



□ AND Gate Truth Table

- Ex. $n=2 \rightarrow \# \text{ of rows } = 2^2 = 4$

Truth Table

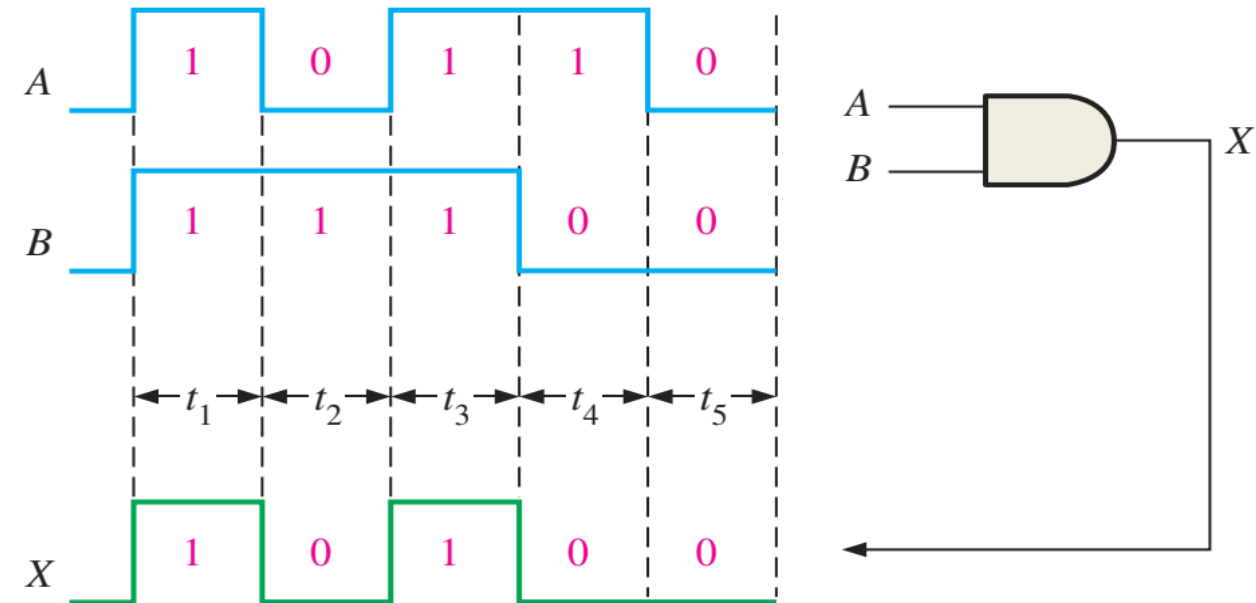
Input		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

The AND Gate (2/3)



□ AND Gate Operation

□ Output will be a **HIGH** only when all the inputs are **HIGH**



Example of AND gate operation

□ AND Gate Logic Expression

□ The Boolean expression for the output of a 2-input AND gate

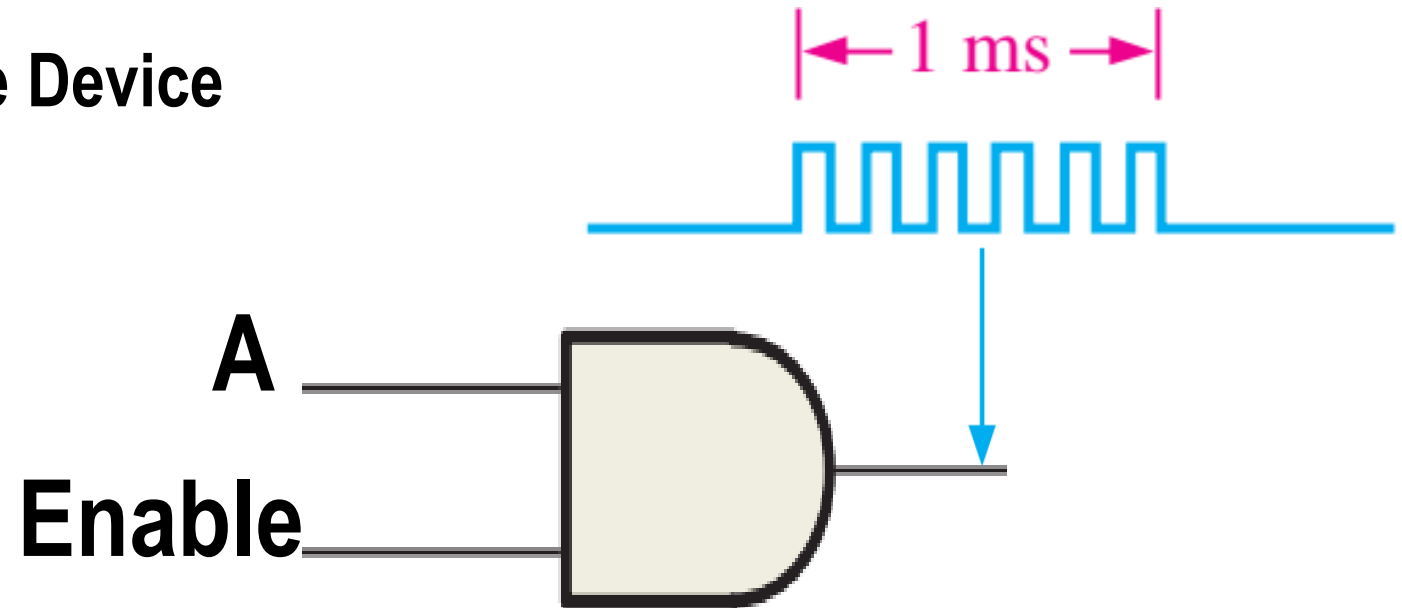
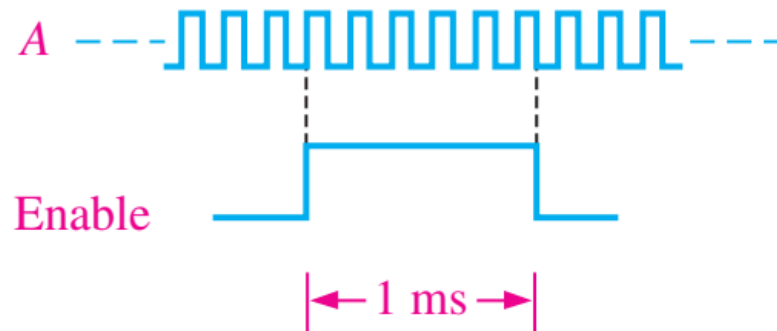
$$X = A \cdot B$$

The AND Gate (3/3)



Application

The AND Gate as an Enable Device



Q1: Build the truth table for a 3-input AND Gate, Determine the total number of possible input combinations, and Draw the symbol.





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The OR Gate (1/3)

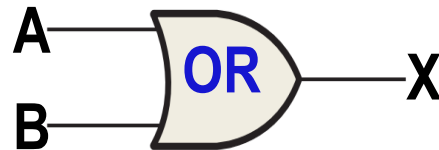


□ Definition

- The **OR** function produces a **HIGH** output when **one or more** inputs are **HIGH**
- Basic **OR** gate is composed of two or more inputs and a single output.

□ Symbol

- Standard logic symbols for the OR



□ OR Gate Truth Table

- Ex. $n=2 \rightarrow \# \text{ of rows } = 2^2 = 4$

Truth Table

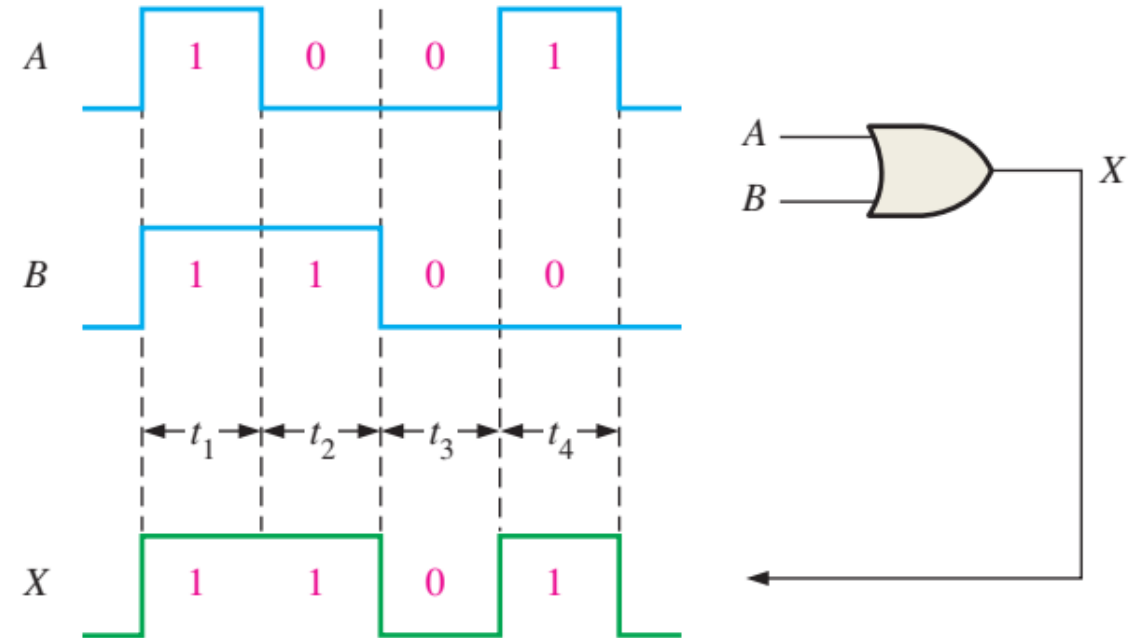
Input		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

The OR Gate (2/3)



□ OR Gate Operation

□ Output will be a **HIGH** when one or more inputs are **HIGH**



Example of OR gate operation

□ OR Gate Logic Expression

□ The Boolean expression for the output of a 2-input OR Gate

$$X = A + B$$

The OR Gate (3/3)

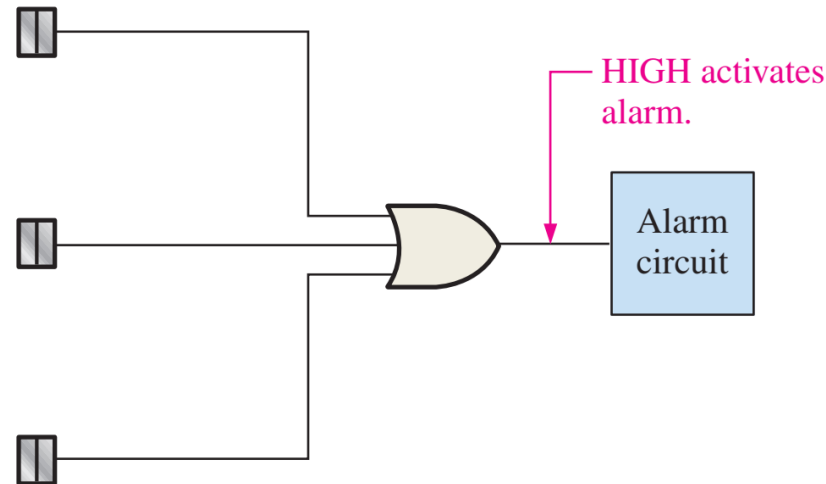


□ Application

□ A simplified intrusion detection system

Open door/window sensors

HIGH = Open
LOW = Closed



Q1: Build the truth table for a 3-input OR Gate, Determine the total number of possible input combinations, and Draw the symbol.



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The NAND Gate (1/2)

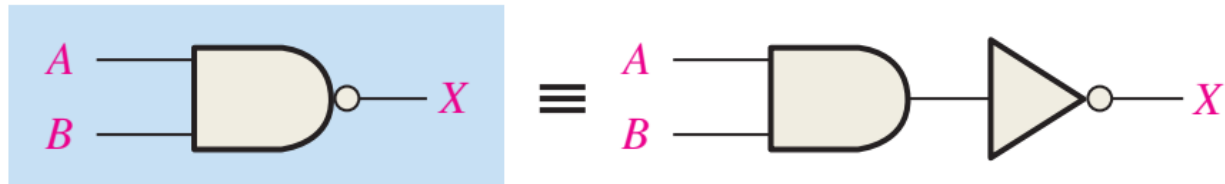


□ Definition

- The term **NAND** is a contraction of NOT-AND → **Universal Gate**
- The **NAND** function produces a **LOW** output only when all the inputs are **HIGH**

□ Symbol

- Standard logic symbols for the NAND



□ NAND Gate Truth Table

- Ex. $n=2 \rightarrow \# \text{ of rows } = 2^2 = 4$

Truth Table

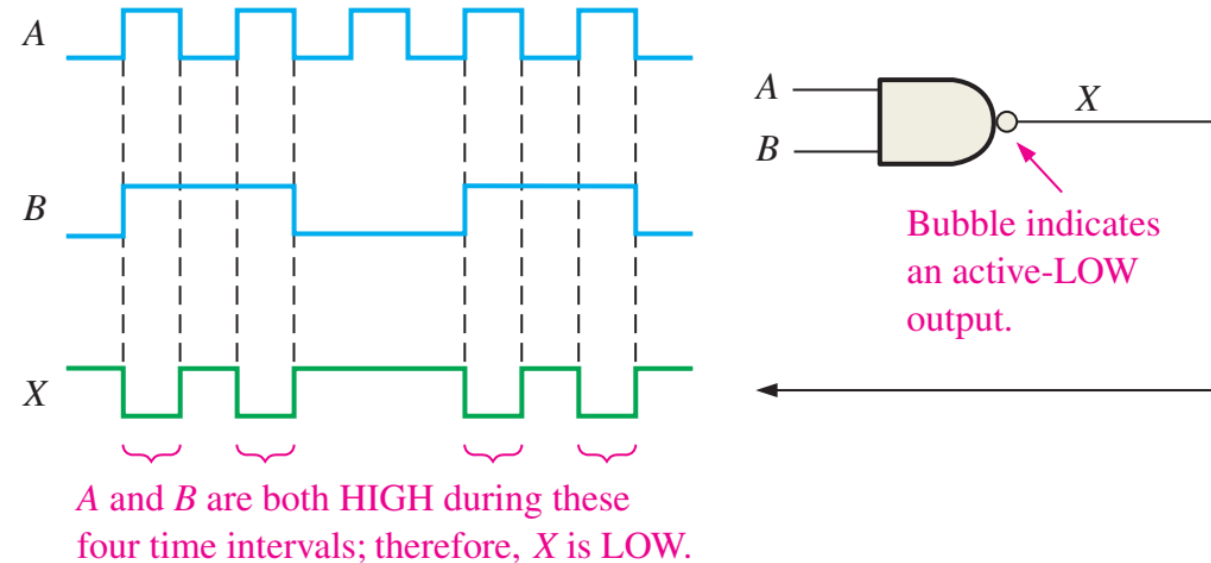
Input		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

The NAND Gate (2/2)



□ NAND Gate Operation

- Output will be a **LOW** only when all the inputs are **HIGH**



Example of NAND Gate operation

□ NAND Gate Logic Expression

- The Boolean expression for the output of a 2-input NAND Gate

$$X = \overline{A \cdot B}$$



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The NOR Gate (1/2)

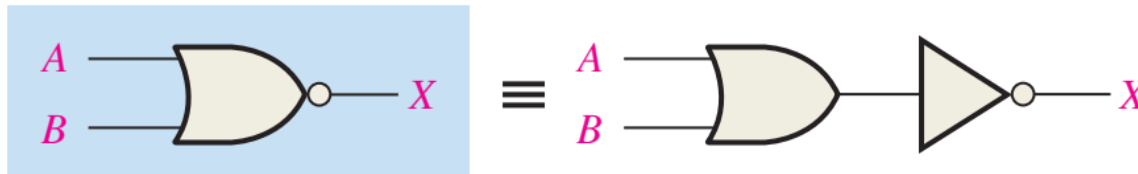


□ Definition

- The term **NOR** is a contraction of NOT-OR → **Universal Gate**
- The **NOR** function produces a **LOW** output when any of its input is **high**

□ Symbol

- Standard logic symbols for the NOR Gate



□ NOR Gate Truth Table

- Ex. $n=2 \rightarrow \# \text{ of rows } = 2^2 = 4$

Truth Table

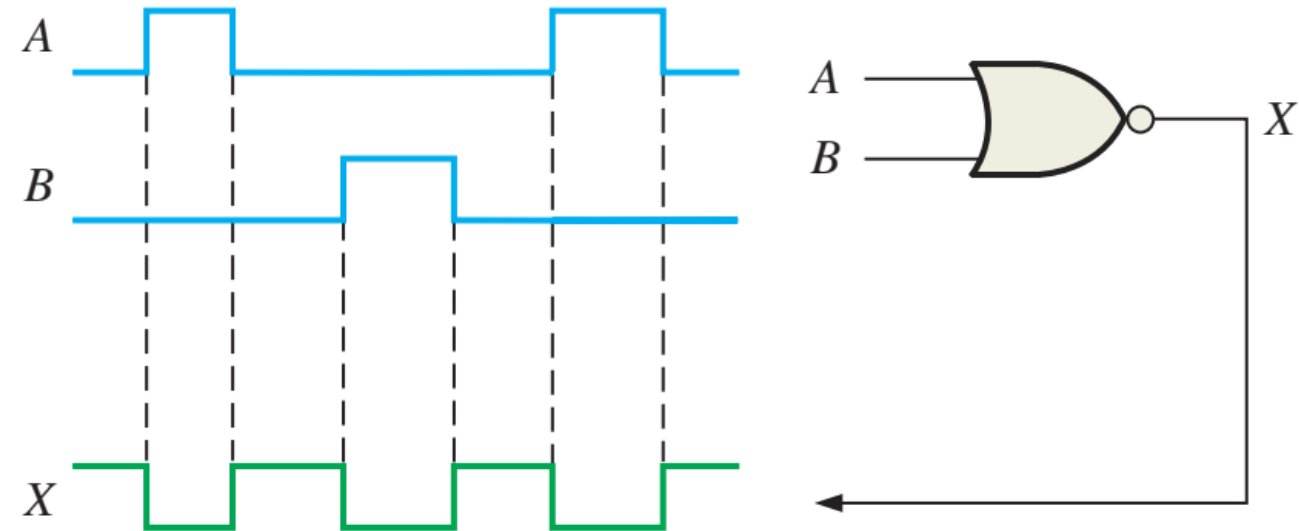
Input		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

The NOR Gate (2/2)



❑ NOR Gate Operation

- ❑ Output will be a **LOW** when any of its input is **High**



Example of NOR gate operation

❑ NOR Gate Logic Expression

- ❑ The Boolean expression for the output of a 2-input NOR Gate

$$X = \overline{A + B}$$



- ☐ The NOT Gate
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- ☐ The OR Gate
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- ☐ The XNOR Gate

The XOR Gate (1/2)



□ Definition

- The term **XOR** is a contraction of Exclusive-OR
- The **XOR** function produces a **High** output when both inputs are different

□ Symbol

- Standard logic symbols for the XOR



□ XOR Gate Truth Table

- Ex. $n=2 \rightarrow \# \text{ of rows } = 2^2 = 4$

Truth Table

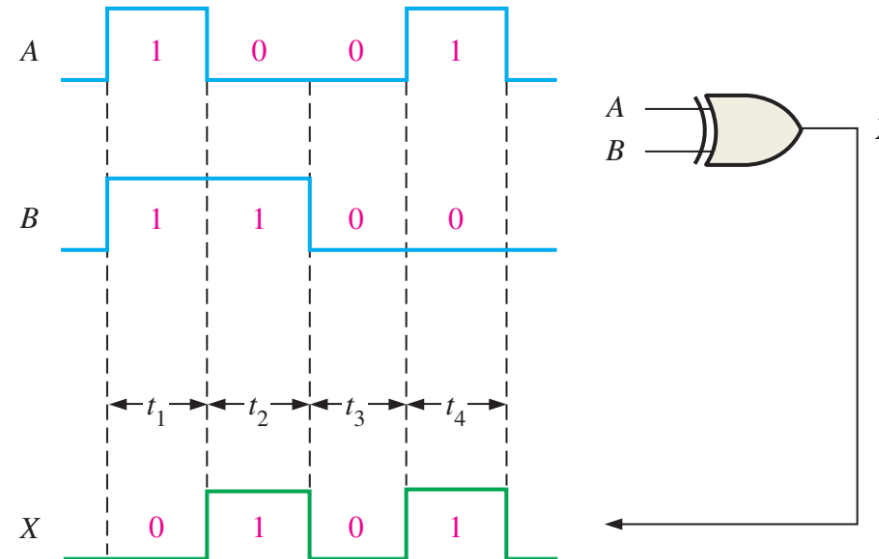
Input		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

The XOR Gate (2/2)



XOR Gate Operation

- Output will be a **High** when both inputs are different



Example of XOR gate operation

XOR Gate Logic Expression

- The Boolean expression for the output of a 2-input XOR Gate

$$X = A \oplus B$$



- ☐ The NOT Gate
- ☐ The AND Gate
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The XNOR Gate (1/2)

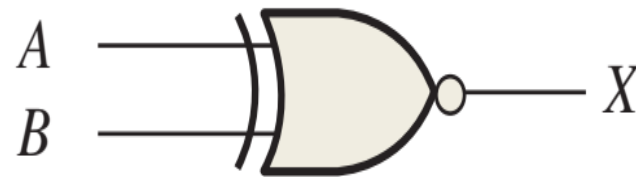


□ Definition

- The term **XNOR** is a contraction of Exclusive-NOR
- The **XNOR** function produces a **High** output when both inputs are the same

□ Symbol

- Standard logic symbols for the XNOR



□ XNOR Gate Truth Table

- Ex. $n=2 \rightarrow \# \text{ of rows } = 2^2 = 4$

Truth Table

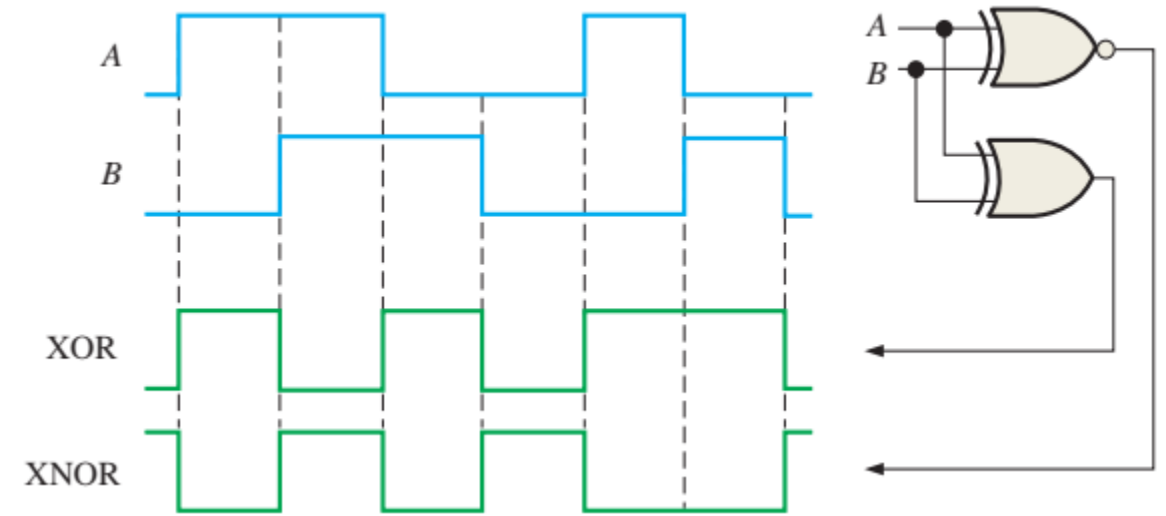
Input		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

The XNOR Gate (2/2)



❑ XNOR Gate Operation

- ❑ Output will be a **High** when both inputs are the same



Example of XOR and XNOR gate operation

❑ XNOR Gate Logic Expression

- ❑ The Boolean expression for the output of a 2-input XNOR gate

$$X = A \odot B$$

Thanks!